H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac															
				Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
				Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lw

Only 15 elements are magnetically ordered in the solid state: <u>d and f elements</u>

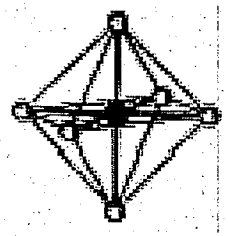
Lecture 8

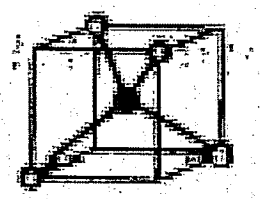
Magnetic ions and atoms in a crystal: crystal field

In a solid: the potential has no spherical symmetry

Cubic crystal: $V_c = V_0(x^4 + y^4 + z^4)$
1-dimensional system: $V_c = V_0 z^2$

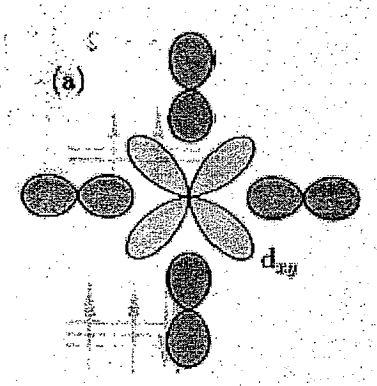
Origin of the crystal field: electrostatic interaction of the electronic charge density with the neighbouring ions

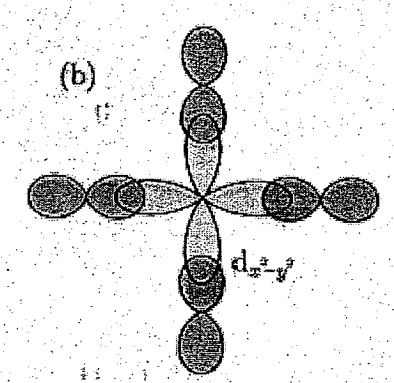




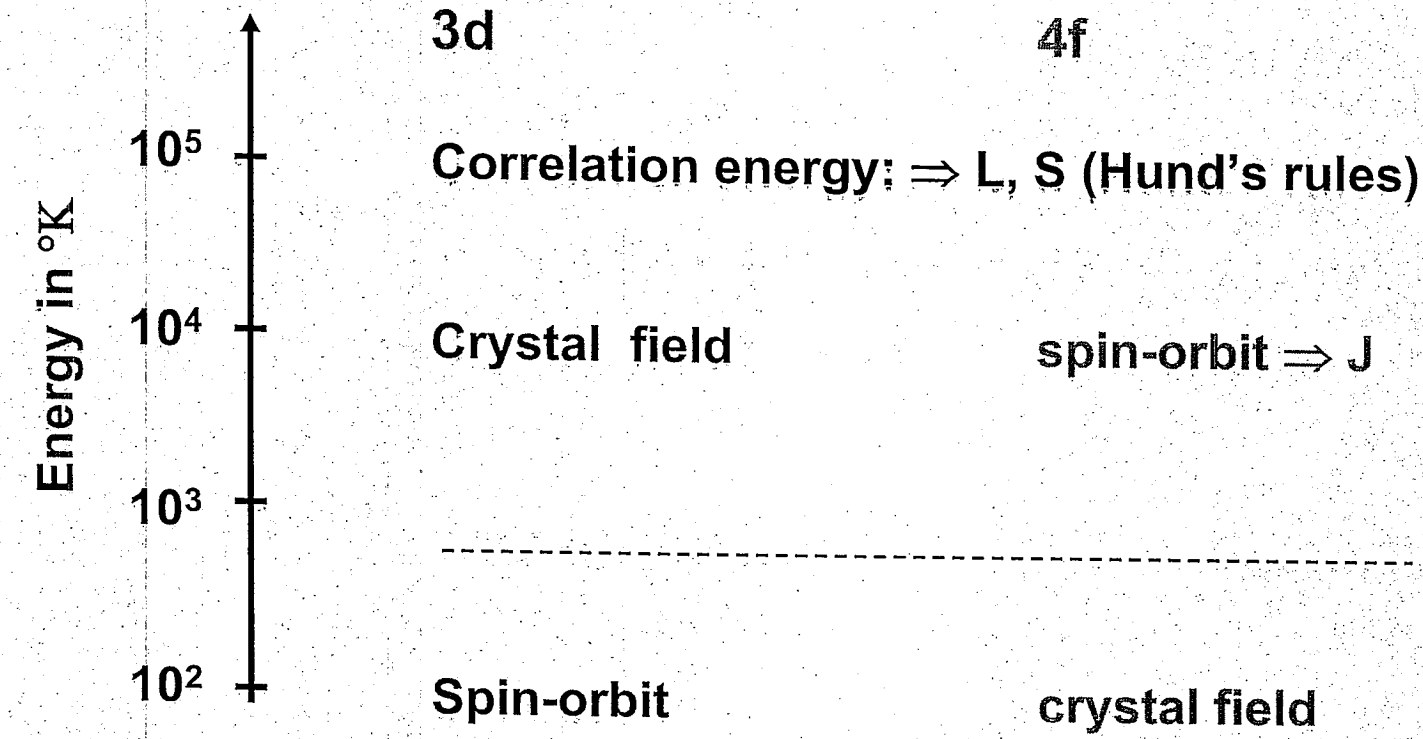
2 cubic environments: octahedral and tetrahedral

In octahedral environment xy orbital is lower in energy than x^2-y^2





Order of magnitude of the energies involved:



-3d electrons less localized than 4f: crystal field larger in 3d than in 4f
-Spin orbit coupling α Z^4: larger in 4f than in 3d
-In 3d ions : crystal field >> spin-orbit: orbital moment is reduced (or quenched)
-In 4f ions: spin-orbit >> crystal field: the ground state multiplet (J) is splitted by crystal field

Crystal field for 4f ions: crystal field is a perturbation on the ground state multiplet defined by J

For a 1-dimensional crystal field: $V_c = BJ_z^2$

J=4, B>0 ($4f^2$: Pr^{3+})

———	± 4
———	±3
———	±2
———	±1
———	Jz =0

Ground state J_z=0

« non- magnetic»

J=5/2, B<0 ($4f^1$: Ce^{3+})

———	± 1/2
———	±3/2
———	± 5/2

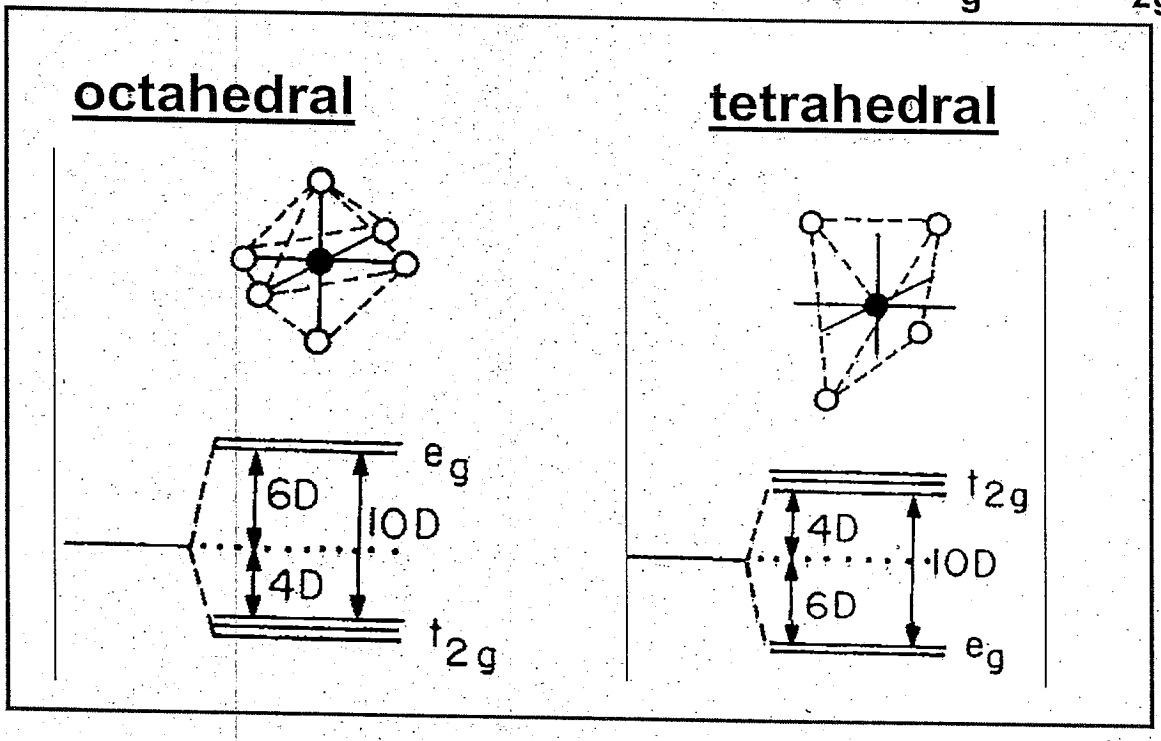
ground state: J_z=± 5/2

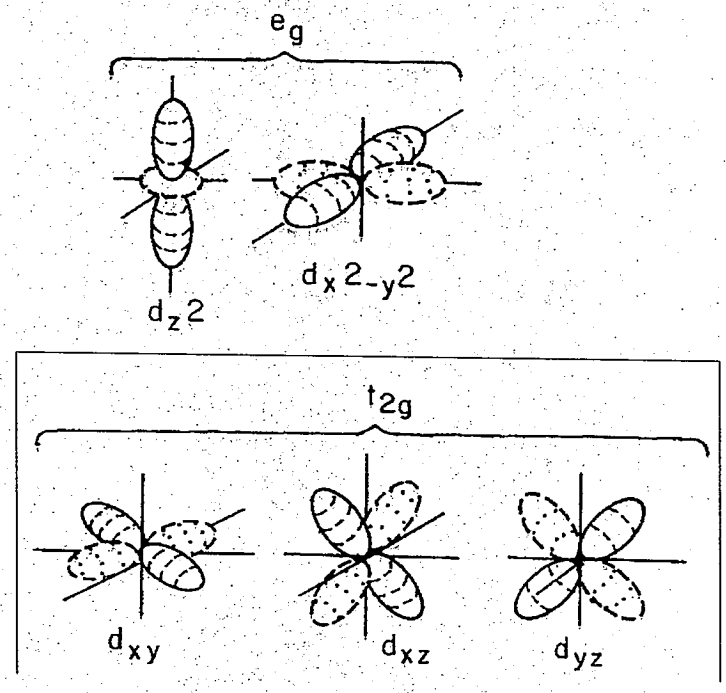
Ising-like

Crystal field for 3d ions : cubic crystal field

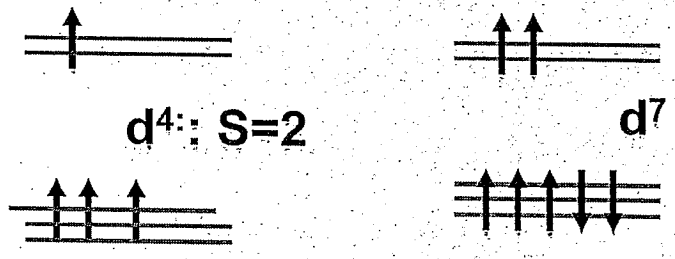
Crystal field potential: $V_c = V_0(x^4 + y^4 + z^4 - 3/5r^4)$

d orbitals are splitted in 2 groups: e_g and t_{2g}; 2 cases:





Filling of the d-orbitals following 1st Hund's rule (S maximum)



d^4: S=2

d^7 : S=3/2

Quenching of orbital magnetic moment:

Wave functions of e_g states: $\frac{1}{\sqrt{2}}(Y_2^2 + Y_2^{-2})$ and Y_2^0

No orbital magnetism:

$$\left\langle e_g^1 \middle| L_\alpha \middle| e_g^2 \right\rangle = 0 \,,\, \alpha = x, y, z$$

t_{2g} states: $\frac{1}{\sqrt{2}}(Y_2^2 - Y_2^{-2}), \frac{1}{\sqrt{2}}(Y_2^1 - Y_2^{-1})$, and $\frac{1}{\sqrt{2}}(Y_2^1 + Y_2^{-1})$

Diagonal matrix elements of L_α vanish, only off-diagonal elements:
$\Rightarrow$ «reduced» orbital moment

In cubic symmetry orbital magnetism is either quenched (e_g) or reduced (t_{2g}): L is not given by Hund's rule

Spin-orbit coupling is a smaller effect, acting mainly in t_{2g} states

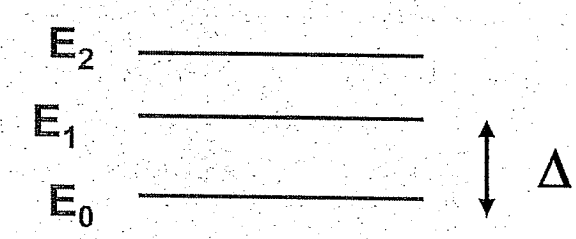
<u>Single ion anisotropy</u>: due to crystal field + spin orbit

<u>3d ion</u>: spin orbit coupling is a perturbation

$$\langle 0|\vec{L}|0\rangle = 0 \Rightarrow \langle 0|\lambda\vec{L}.\vec{S}|0\rangle = 0 \Rightarrow \text{no effect to 1st order}$$

2nd order perturbation:

$$\Delta E = -\sum \frac{\langle 0|\lambda\vec{L}\vec{S}|n\rangle\langle n|\lambda\vec{L}\vec{S}|0\rangle}{E_n - E_0} = -\sum_{\alpha\beta} \Lambda_{\alpha\beta} S_\alpha S_\beta \qquad \Delta E \approx \frac{\lambda^2}{\Delta}$$



Higher order terms in high symmetry environment (cubic): $\Delta E \propto \frac{\lambda^4}{\Delta^3}$

Energy depends on the direction of magnetization:

i.e. $-DS_z^2$ with $D > 0$ or < 0

<u>Rare earth ions</u>: similar form $\quad \Delta E = \sum_{\alpha\beta} \Lambda_{\alpha\beta} J_\alpha J_\beta$

Order of magnitude: for cubic 3d metals: a few °K

for rare earth: 10^2 to 10^3 °K

<u>High spin- low spin states</u> (transition metals)

Example of Fe^{2+} (d^6 configuration) in a cubic crystal field

-If small crystal field, Hund's rule $\Rightarrow$ S=2 (high spin)

e_g ↑↑

t_{2g} ↑↑↑↓

-If large crystal field, all t_{2g} states are filled $\Rightarrow$ S=0 (low spin)

e_g ——— ———

t_{2g} ↑↑↑↓↓↓

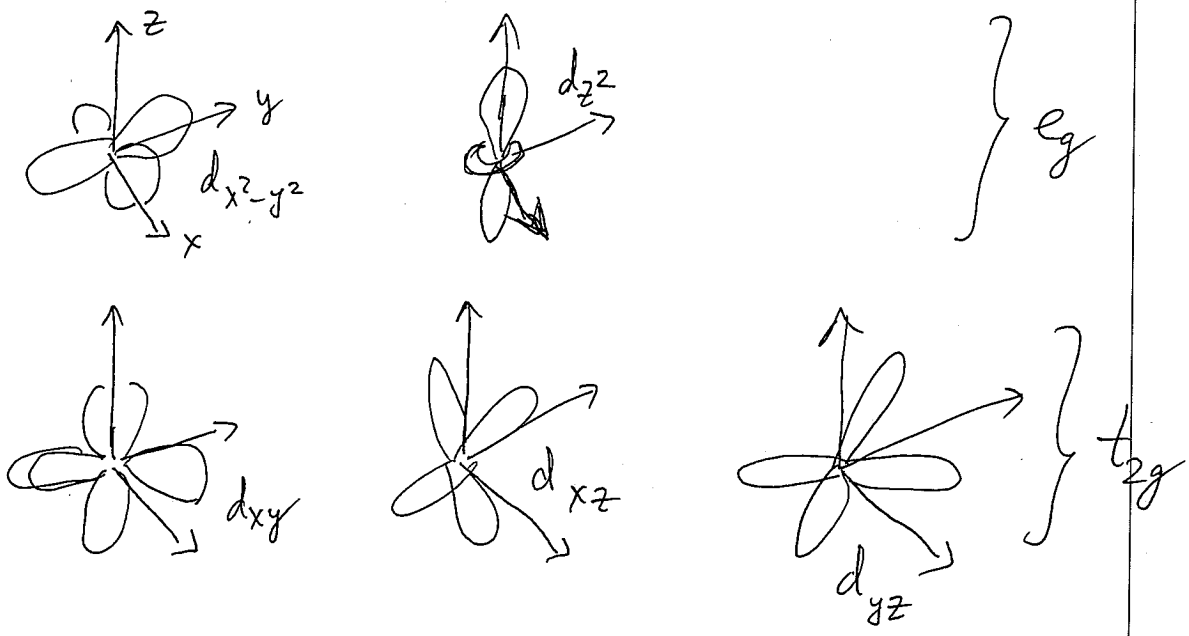
1st Hund's rule is violated!

Crystal fields

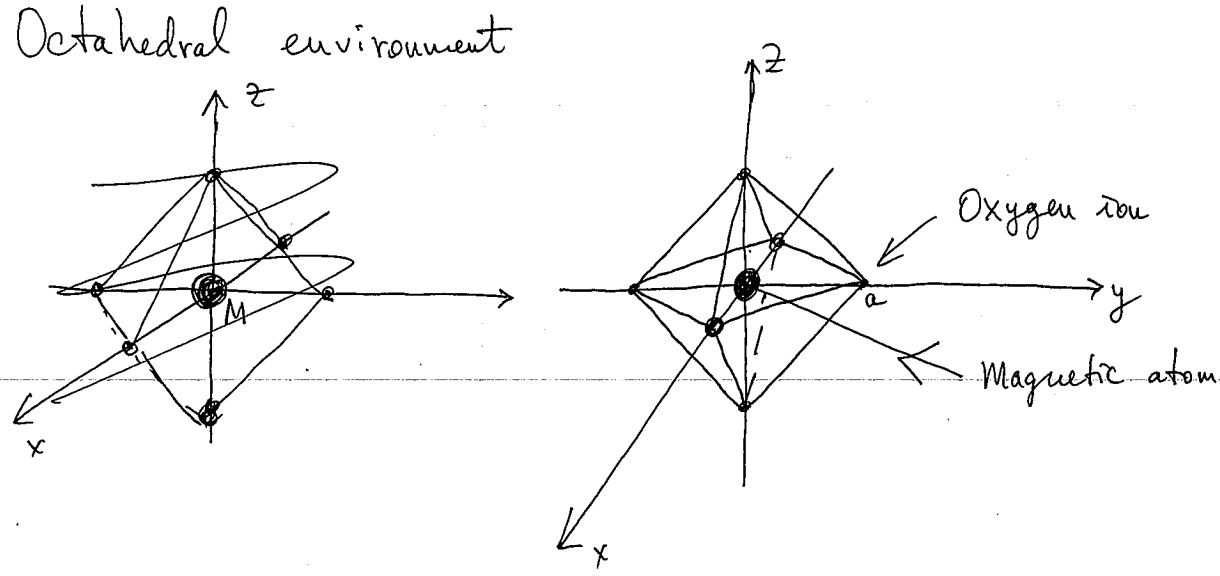
We now describe effect of local environment of magnetic atom (ion) in a crystal.

Wavefunctions of $L \neq 0$ states have rather complicated angular structure, which is sensitive to the local electric potential. This effect for ions of 3d group ($3d^1 - 3d^{10}$) where it competes with Hund's rules (in particular, rule #3) and leads to <u>quenching</u> of orbital momentum

<u>d-orbitals</u>



Octahedral environment

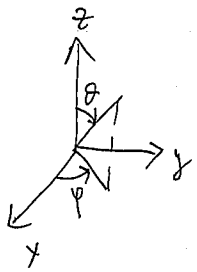


D orbitals ($l=2$)

are linear combinations of spherical harmonics

$$Y_{xy} = \frac{1}{i\sqrt{2}}\left(Y_2^2 - Y_2^{-2}\right) = \frac{1}{i\sqrt{2}}\left(\frac{15}{32\pi}\right)^{1/2} \sin^2\theta \underbrace{\left(e^{i2\varphi} - e^{-i2\varphi}\right)}_{i\sin 2\varphi} =$$

$$= \left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta \sin 2\varphi = \left(\frac{15}{4\pi}\right)^{1/2} \frac{xy}{r^2}$$



$$Y_{x^2-y^2} = \frac{1}{\sqrt{2}}\left(Y_2^2 + Y_2^{-2}\right) = \left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta \cos 2\varphi = \left(\frac{15}{16\pi}\right)^{1/2} \frac{x^2-y^2}{r^2}$$

$$Y_{yz} = \frac{-1}{i\sqrt{2}}\left(Y_2^1 + Y_2^{-1}\right) = \frac{-1}{i\sqrt{2}}\left(\frac{15}{8\pi}\right)^{1/2} \sin\theta\cos\theta\left(-e^{+i\varphi} + e^{-i\varphi}\right) =$$

$$= \left(\frac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta \sin\varphi = \left(\frac{15}{4\pi}\right)^{1/2} \frac{yz}{r^2}$$

$$Y_{zx} = \frac{1}{\sqrt{2}}\left(-Y_2^1 + Y_2^{-1}\right) = \frac{1}{\sqrt{2}}\left(\frac{15}{8\pi}\right)^{1/2} \sin\theta\cos\theta\left(e^{i\varphi} + e^{-i\varphi}\right)$$

$$= \left(\frac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\cos\varphi = \left(\frac{15}{4\pi}\right)^{1/2} \frac{xz}{r^2}$$

$$Y_{z^2} = Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2}\left(3\cos^2\theta - 1\right) = \left(\frac{5}{16\pi}\right)^{1/2} \frac{3z^2 - r^2}{r^2} =$$

$$= \left(\frac{15}{16\pi}\right)^{1/2} \frac{2z^2 - x^2 - y^2}{r^2}$$

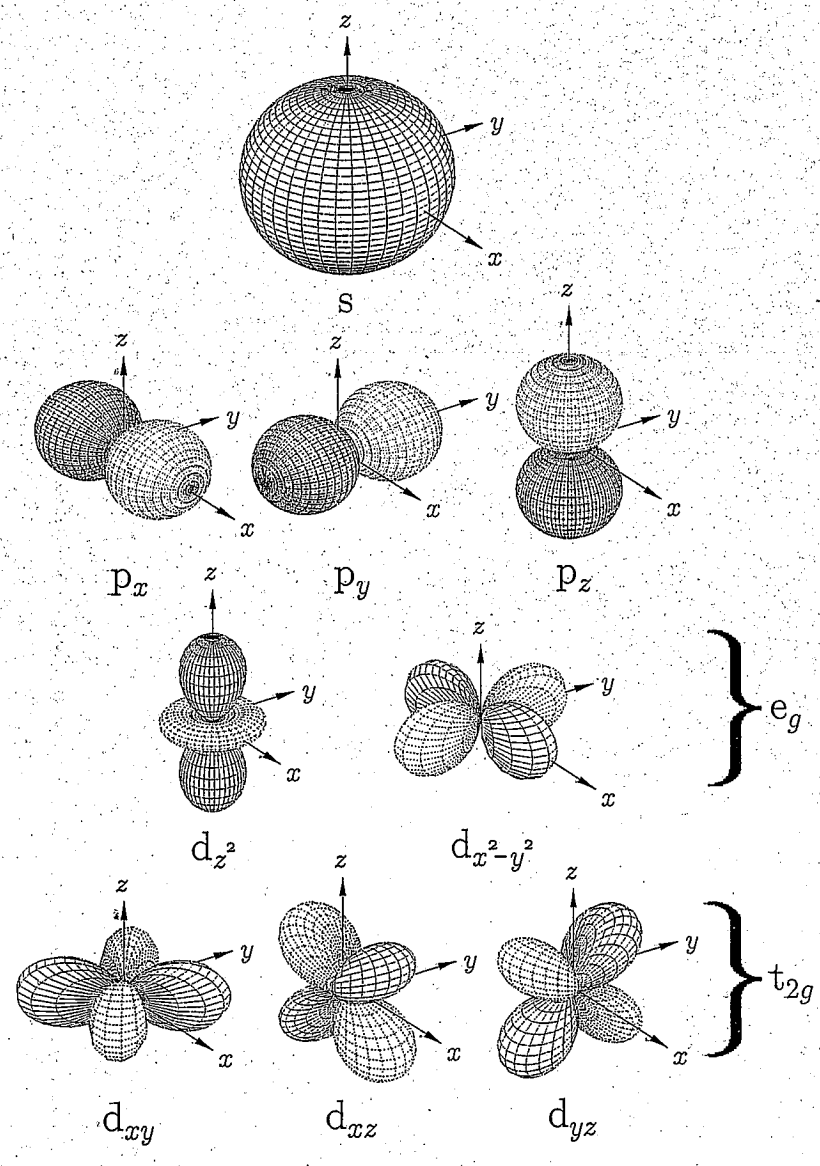


Fig. 3.1 The angular distribution of the s, p and d orbitals. The d_{z^2} and $d_{x^2-y^2}$ levels are grouped together and called the e_g levels. The d_{xy}, d_{xz} and d_{yz} levels are grouped together and called the t_{2g} levels. The d_{z^2} orbital is sometimes referred to as $d_{3z^2-r^2}$.

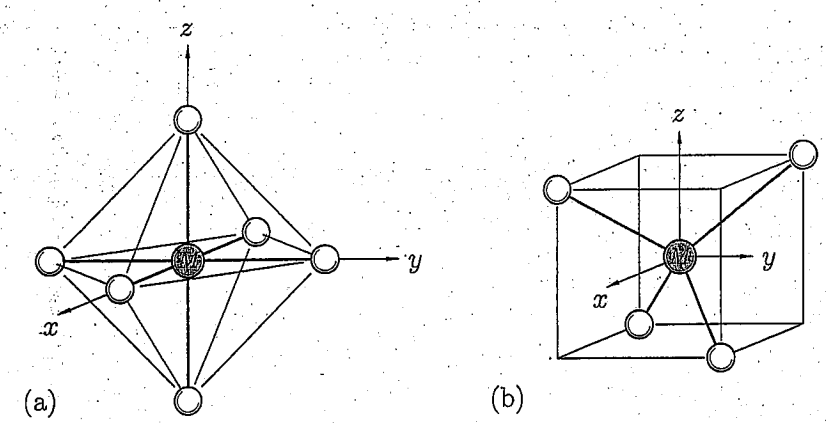
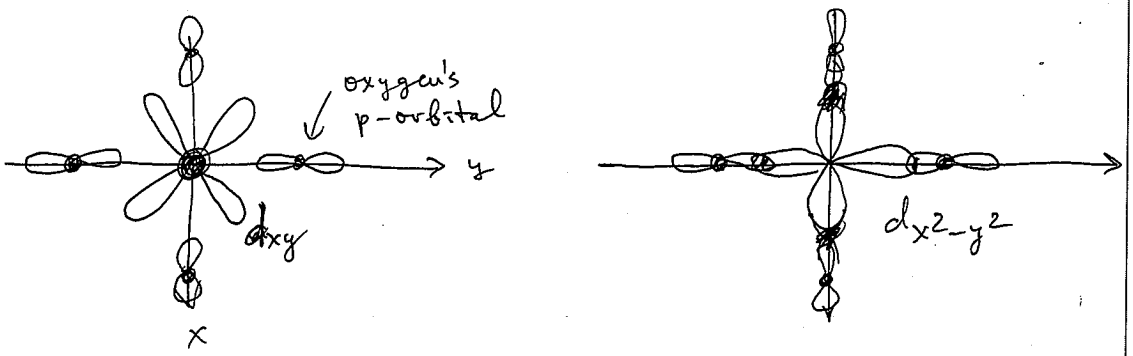


Fig. 3.2 A metal atom M in an (a) octahedral and (b) tetrahedral environment. The octahedral environment is found in many transition metal oxides where oxygen anions sit on the corners of an octahedron with the metal atom at the centre. The tetrahedral environment can be conveniently described by considering alternate corners of a cube as shown.

Top view explains the basic feature best:



d_{xy} overlaps less that $d_{x^2-y^2}$ $\Rightarrow$ energies of these orbitals must be different!

Consider electric potential of 6 charges in corners of an octahedron : $(\pm a, 0, 0)$, $(0, \pm a, 0)$ and $(0, 0, \pm a)$.

$V(r) = \dfrac{q}{4\pi\varepsilon_0 r}$ where r is measured from the center of octahedron.

$$\Rightarrow V(r) = \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{\sqrt{(a\pm x)^2+y^2+z^2}} + \frac{1}{\sqrt{x^2+(a\pm y)^2+z^2}} + \frac{1}{\sqrt{x^2+y^2+(a\pm z)^2}}\right\}$$

Start with $\tilde{V}_x = \dfrac{1}{\sqrt{(a+x)^2+y^2+z^2}} + \dfrac{1}{\sqrt{(a-x)^2+y^2+z^2}} = (x, y, z \ll a) =$

$$= \frac{1}{a}\left\{\frac{1}{\left(1+\frac{2x}{a}+\frac{x^2+y^2+z^2}{a^2}\right)^{1/2}} + \frac{1}{\left(1-\frac{2x}{a}+\frac{x^2+y^2+z^2}{a^2}\right)^{1/2}}\right\} =$$

$$= \frac{1}{a}\left\{\frac{1}{\left(1+\frac{2x}{a}+\frac{r^2}{a^2}\right)^{1/2}} + \frac{1}{\left(1-\frac{2x}{a}+\frac{r^2}{a^2}\right)^{1/2}}\right\}$$

$$V(r) = \frac{q}{4\pi\varepsilon_0 a}\left\{\frac{1}{\sqrt{1+\frac{2x}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1-\frac{2x}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1+\frac{2y}{a}+\frac{r^2}{a^2}}} + \right.$$

$$\left. + \frac{1}{\sqrt{1-\frac{2y}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1+\frac{2z}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1-\frac{2z}{a}+\frac{r^2}{a^2}}}\right\}$$

Mathematica

```
In[16]:= Series[1/Sqrt[1 + 2 x + x^2 + y^2 + z^2] + 1/Sqrt[1 - 2 x + x^2 + y^2 + z^2] +
    1/Sqrt[1 + 2 y + x^2 + y^2 + z^2] + 1/Sqrt[1 - 2 y + x^2 + y^2 + z^2] +
    1/Sqrt[1 + 2 z + x^2 + y^2 + z^2] + 1/Sqrt[1 - 2 z + x^2 + y^2 + z^2],
   {x, 0, 4}, {y, 0, 4}, {z, 0, 4}] // FullSimplify[#] & // Normal // Expand
```

Out[16]=

$$6 + \frac{7x^4}{2} - \frac{21x^2y^2}{2} - \cancel{\frac{45x^4y^2}{8}} + \frac{7y^4}{2} - \cancel{\frac{45x^2y^4}{8}} + \cancel{\frac{3465x^4y^4}{32}} - \frac{21x^2z^2}{2} - \cancel{\frac{45x^4z^2}{8}} - \frac{21y^2z^2}{2} + \cancel{\frac{135}{2}x^2y^2z^2} - \cancel{\frac{45y^4z^2}{8}} - \cancel{\frac{20475}{32}x^4y^4z^2} + \frac{7z^4}{2} - \cancel{\frac{45x^2z^4}{8}} + \cancel{\frac{3465x^4z^4}{32}} - \cancel{\frac{45y^2z^4}{8}} - \cancel{\frac{20475}{32}x^4y^2z^4} + \cancel{\frac{3465y^4z^4}{32}} - \cancel{\frac{20475}{32}x^2y^4z^4} + \cancel{\frac{155925}{32}x^4y^4z^4} =$$

$$\Rightarrow \tilde{V} = 6 + \frac{7}{2}\left(x^4+y^4+z^4\right) - \frac{21}{2}\left(x^2y^2+x^2z^2+y^2z^2\right) + O(x^6)$$

Observe that $\left(x^2+y^2+z^2\right)^2 \equiv \left(r^2\right)^2 = x^4+y^4+z^4+2x^2y^2+2x^2z^2+2y^2z^2$

$$\Rightarrow \cancel{x^2+y^2} \quad x^2y^2+x^2z^2+y^2z^2 = \frac{1}{2}\left(r^4 - x^4-y^4-z^4\right)$$

$$\tilde{V} = 6 + \frac{7}{2}\left(x^4+y^4+z^4\right) - \frac{21}{2}\,\frac{1}{2}\left(r^4 - \left(x^4+y^4+z^4\right)\right) =$$

$$= 6 + \underbrace{\left(\frac{7}{2}+\frac{21}{4}\right)}_{\frac{35}{4}}\left(x^4+y^4+z^4\right) - \frac{21}{4}r^4 =$$

$$= 6 + \frac{35}{4}\left(x^4+y^4+z^4 - \frac{21}{35}r^4\right) = 6 + \frac{35}{4}\left(x^4+y^4+z^4 - \frac{3}{5}r^4\right) + O(r^6).$$

$$\frac{21}{35} = \frac{3}{5}$$

Mathematica shows that

$$V(r) = \frac{q}{4\pi\varepsilon_0 a}\left[6 + \frac{35}{4a^4}\left(x^4+y^4+z^4 - \frac{3}{5}r^4\right) + O\left(\frac{r^6}{a^6}\right)\right]$$

This is crystal field potential produced by 6 anions (oxygens) of charge q each. Electron on magnetic ion M experiences this $V(r)$.

Drop constant and write $V(r) = D\left(x^4+y^4+z^4 - \frac{3}{5}r^4\right)$.

Consider $V(r)$ a perturbation on degenerate d-levels of M-ion ($L=2 \Rightarrow 5$ degenerate levels).

Moreover, $x^4+y^4+z^4 = r^4\left[(\sin\theta\cos\varphi)^4 + (\sin\theta\sin\varphi)^4 + (\cos\theta)^4\right] =$

$$= r^4\left[\sin^4\theta\,\underbrace{(\cos^4\varphi + \sin^4\varphi)}_{\frac{1}{8}(e^{i4\varphi}+e^{-i4\varphi}+6)} + \cos^4\theta\right] = \frac{r^4}{8}\left[\sin^4\theta\,(6+e^{i4\varphi}+e^{-i4\varphi}) + 8\cos^4\theta\right]$$

$$\Rightarrow V(r) = \frac{Dr^4}{8}\left\{\sin^4\theta\,(6+e^{i4\varphi}+e^{-i4\varphi}) + 8\cos^4\theta - \frac{24}{5}\right\}$$

Wave functions of 3d orbitals: $n=3, l=2, m=0,\pm1,\pm2$

$$\psi_{nlm}(r,\theta,\varphi) = R_{32}(r)\,Y_2^m(\theta,\varphi)$$

$$Y_2^{\pm2} = \sqrt{\frac{15}{32\pi}}\,\sin^2\theta\,e^{\pm i2\varphi}$$

$$Y_2^{\pm1} = \mp\sqrt{\frac{15}{32\pi}\cdot 4}\,\sin\theta\,e^{\pm i\varphi} = \sqrt{\frac{15}{32\pi}}\left(\mp 2\sin\theta\,e^{\pm i\varphi}\right)$$

$$Y_2^0 = \sqrt{\frac{15}{32\pi}\cdot\frac{2}{3}}\,(3\cos^2\theta - 1) = \sqrt{\frac{15}{32\pi}}\left(\sqrt{\frac{2}{3}}\,[3\cos^2\theta - 1]\right)$$

* Absorb normalization factor $\sqrt{\frac{15}{32\pi}}$ into $R_{32}(r)$; $R \to \tilde{R}$

$$\psi_{32m}(r,\theta,\varphi) = \hat{R}_{32}(r) \begin{cases} \sin^2\theta\, e^{\pm i2\varphi} \\ \mp 2\sin\theta\cos\theta\, e^{\pm i\varphi} \\ \sqrt{\frac{2}{3}}\,(3\cos^2\theta - 1) \end{cases} = \begin{cases} |\pm 2\rangle \\ |\pm 1\rangle \\ |0\rangle \end{cases}$$

where $|\pm m\rangle$ denotes L_z eigenstates.

General state $|\psi\rangle = \sum_{j=-2}^{2} a_j |j\rangle \rightarrow \begin{pmatrix} a_2 \\ a_1 \\ a_0 \\ a_{-1} \\ a_{-2} \end{pmatrix}$

Find $\langle\psi|V(r)|\psi\rangle$ in this 5-dimensional basis.

It is easy to see that $\langle i|V|j\rangle = 0$ with $i \neq j$ (<u>for $i \neq \pm 2, j \neq \pm 2$</u>).

Indeed $V = A(e^{i4\varphi} + e^{-i4\varphi}) + B$ while $|i\rangle \sim e^{im_i\varphi}$

$$\Rightarrow \langle i|V|j\rangle \sim \int_0^{2\pi} e^{-im_i\varphi}\left[A(e^{i4\varphi} + e^{-i4\varphi}) + B\right] e^{-im_j\varphi}\, d\varphi$$

For $m_i \neq m_j$ $\int d\varphi\, e^{i(m_j - m_i)\varphi} = 0$ and $\int d\varphi\, e^{i(m_j - m_i \pm 4)\varphi} = 0$

<u>unless</u> $m_j - m_i = \pm 4$ when the exponent $= 0$

$$\Rightarrow \langle i|V|j\rangle \neq 0 \text{ for } \left[\begin{array}{l} i = 2,\ j = -2 \\ i = -2,\ j = 2 \end{array}\right.$$

Clearly $\langle 2|V|-2\rangle = \langle -2|V|2\rangle = V_{2,-2}$

$$V_{2,-2} = \int_0^\infty dr \cdot r^2\, r^4 \left(\frac{D}{8}\right)\tilde{R}_{32}^2 \int_0^\pi d\theta \sin\theta \int d\varphi \sin^2\theta\, e^{-i2\varphi}\left\{\sin^4\theta\,(6 + e^{i4\varphi} + e^{-i4\varphi}) + 8\cos^4\theta - \frac{24}{5}\right\} \sin^2\theta\, e^{-i2\varphi} = \int_0^\infty dr\, r^6\, \tilde{R}_{32}^2 \left(\frac{D}{8}\right) \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \sin^8\theta.$$

Diagonal components $V_{jj} = \langle j|V|j\rangle \neq 0$ and there we can remove $e^{\pm i4\varphi}$ terms from $V(r)$ altogether: $V(r) \rightarrow \frac{Dr^4}{8}\left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$

$$V_{2,2} = V_{-2,-2} = \int dr\, r^6\, \tilde{R}_{32}^2(r)\, \frac{D}{8} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \sin^4\theta \left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$$

$$V_{1,1} = V_{-1,-1} = \int dr\, r^6\, \tilde{R}^2_{32}(r)\, \frac{D}{8} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta\; 4\sin^2\theta\, \cos^2\theta \left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$$

$$V_{0,0} = \int dr\, r^6\, \tilde{R}^2_{32}(r)\, \frac{D}{8} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \left(\frac{2}{3}\right)\left(3\cos^2\theta - 1\right)^2 \left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$$

Denoting $\alpha \equiv \int_0^{\infty} dr\, r^6\, \tilde{R}^2_{32}(r)\, \frac{D}{8}$, we have (using Mathematica)

$$V_{2,2} = V_{-2,-2} = \alpha\,(2\pi)\,\frac{256}{1575}$$

$$V_{1,1} = V_{-1,-1} = \alpha\,(2\pi)\left(-\frac{1024}{1575}\right)$$

$$V_{0,0} = \alpha\,(2\pi)\,\frac{512}{525} = \alpha\,(2\pi)\,\frac{3\cdot 512}{1575} = \alpha\,(2\pi)\,\frac{1536}{1575}$$

$$V_{2,-2} = \alpha\,(2\pi)\,\frac{256}{315} = \alpha\,(2\pi)\,\frac{5\cdot 256}{1575} = \alpha\,(2\pi)\,\frac{1280}{1575}$$

Denoting now $V_{2,2} = (2\pi)(\alpha)\left(\frac{256}{1575}\right) \equiv A$, we express all other elements as

$$V_{1,1} = -4A \;;\quad V_{0,0} = 6A \;;\quad V_{2,-2} = 5A$$

This remarkable symmetry hints at some bigger picture

$$\Rightarrow \quad V = \begin{pmatrix} A & 0 & 0 & 0 & 5A \\ 0 & -4A & 0 & 0 & 0 \\ 0 & 0 & 6A & 0 & 0 \\ 0 & 0 & 0 & -4A & 0 \\ 5A & 0 & 0 & 0 & A \end{pmatrix} = A \begin{pmatrix} 1 & 0 & 0 & 0 & 5 \\ 0 & -4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & -4 & 0 \\ 5 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The eigenvalues are: $\lambda_1 = 6A$ (2-fold degenerate)

$\lambda_2 = -4A$ (3-fold)

with eigenvectors:

eigenvectors	$\lvert 0\rangle$	$\frac{\lvert 2\rangle + \lvert -2\rangle}{\sqrt{2}}$	$\lvert 1\rangle$	$\lvert -1\rangle$	$\frac{\lvert 2\rangle - \lvert -2\rangle}{\sqrt{2}}$
eigenvalues	$6A$	$6A$	$-4A$	$-4A$	$-4A$

In magnetic field, levels are in addition split by

$$\mu_B B m_\ell \Rightarrow \delta V = \mathrm{diag}(2\mu_B B,\ 1\mu_B B,\ 0,\ -\mu_B B,\ -2\mu_B B)$$

$$\Rightarrow V + \delta V = \begin{pmatrix} A+2h & 0 & 0 & 0 & 5A \\ 0 & -4A+h & 0 & 0 & 0 \\ 0 & 0 & 6A & 0 & 0 \\ 0 & 0 & 0 & -4A-h & 0 \\ 5A & 0 & 0 & 0 & A-2h \end{pmatrix}$$

where for brevity $h \equiv \mu_B B$.

Eigenvalues: $\det(V + \delta V - \lambda \mathbb{1}) = 0$

$$= \begin{vmatrix} A+2h-\lambda & 0 & 0 & 0 & 5A \\ 0 & -4A+h-\lambda & 0 & 0 & 0 \\ 0 & 0 & 6A-\lambda & 0 & 0 \\ 0 & 0 & 0 & -4A-h-\lambda & 0 \\ 5A & 0 & 0 & 0 & A-2h-\lambda \end{vmatrix} = (A+2h-\lambda)\left[(4A+\lambda)^2 - h^2\right](6A-\lambda) \cdot (A-2h-\lambda) +$$

$$+ 5A \underbrace{\begin{vmatrix} 0 & -4A+h-\lambda & 0 & 0 \\ 0 & 0 & 6A-\lambda & 0 \\ 0 & 0 & 0 & -4A-h-\lambda \\ 5A & 0 & 0 & 0 \end{vmatrix}}_{-5A\left[(4A+\lambda)^2 - h^2\right](6A-\lambda)} = \left[(4A+\lambda)^2 - h^2\right]\left\{\left[(A-\lambda)^2 - 4h^2\right](6A-\lambda) - 25A^2(6A-\lambda)\right\} =$$

$$= (6A-\lambda)\left[(4A+\lambda)^2 - h^2\right]\left\{(A-\lambda)^2 - 4h^2 - 25A^2\right\}$$

$$(A-\lambda)^2 = 4h^2 + 25A^2$$

$$\lambda = A \pm \sqrt{25A^2 + 4h^2} = A \pm 5A\left(1 + \frac{2h^2}{25A^2}\right) =$$

$$= \begin{cases} 6A + \dfrac{2h^2}{5A} \\ -4A - \dfrac{2h^2}{5A} \end{cases}$$

levels are:

$$\left.\begin{cases} 6A + \dfrac{2\mu_B^2 B^2}{5A} \approx 6A \\ 6A \end{cases}\right\} \text{no splitting, } L=0$$

$$\left.\begin{cases} -4A + \mu_B B \\ -4A - \mu_B B \\ -4A - \dfrac{2\mu_B^2 B^2}{5A} \approx -4A \end{cases}\right\} L=1 \text{ splitting by } B$$

$\Rightarrow$ original $L=2$ is quenched by crystal field to $\underline{L=1}$ (for $A>0$)
at low $T \ll A$

or to $\underline{\underline{L=0}}$ (for $A<0$)

* Can be easily confirmed by calculating susceptibility from partition function for single ion.

$\chi(T \ll A;\ B \to 0)$.

Note that same susceptibility at $\underline{T \gg A}$ is that of $L=2$ ion! (because than CF quenching does not play a role)

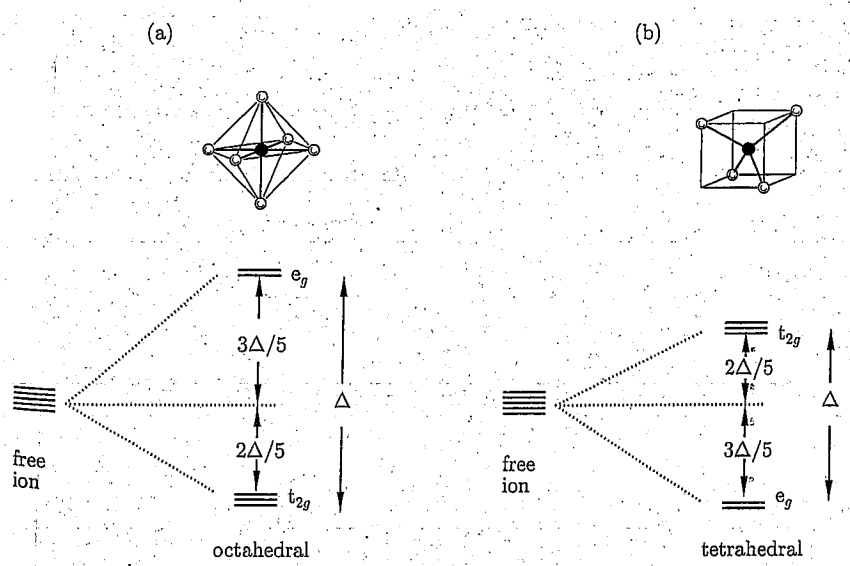


Fig. 3.4 The crystal field in an (a) octahedral and (b) tetrahedral environment.

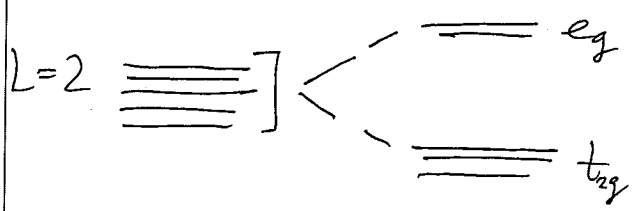
Orbital quenching

→ strong suppression of the angular momentum $\vec{L}$ by the crystal field (CF).
Occurs when CF ≫ Spin-Orbit and results in violation of 3rd Hund's rule (that $J = |L-S| \to L+S$)

→ Pronounced in 3d-ions (Table 3.1 of Blundell) because their $L=2$ d-orbitals "stick out" far from the center and "feel" CF.

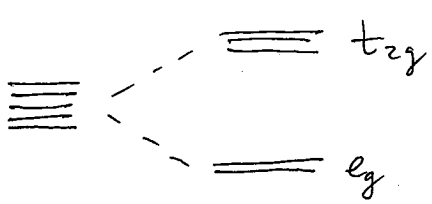
We saw, for <u>octahedral</u> symmetry, that

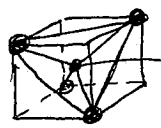
5-fold degenerate



Clearly t_{2g} can be at most $L=1$ state as it is 3-fold degenerate only (d_{xy}, d_{xz}, d_{yz} are related by $\pi/2$ rotations about crystal axes)

for <u>tetrahedral</u>





→ M in the center of tetrahedron

← that would be $L=0$ state because $L \neq$ half-integer: d_{z^2} can not be transformed into $d_{x^2-y^2}$ by discrete rotation consistent with tetrahedron.

More generally, if the ground state is <u>non-degenerate</u>

⇒ w.f. can be choosen real, $|\psi\rangle$. (e.g. $d_{xy}, d_{yz}, d_{xz}, d_{z^2}, d_{x^2-y^2}$ are <u>Real</u>)

$\vec{L}$ is hermitian ⇒ $\langle\psi|\vec{L}|\psi\rangle$ = real.

but $\vec{L} = -\frac{i}{\hbar}\vec{r}\times\vec{\nabla}$ is imaginary

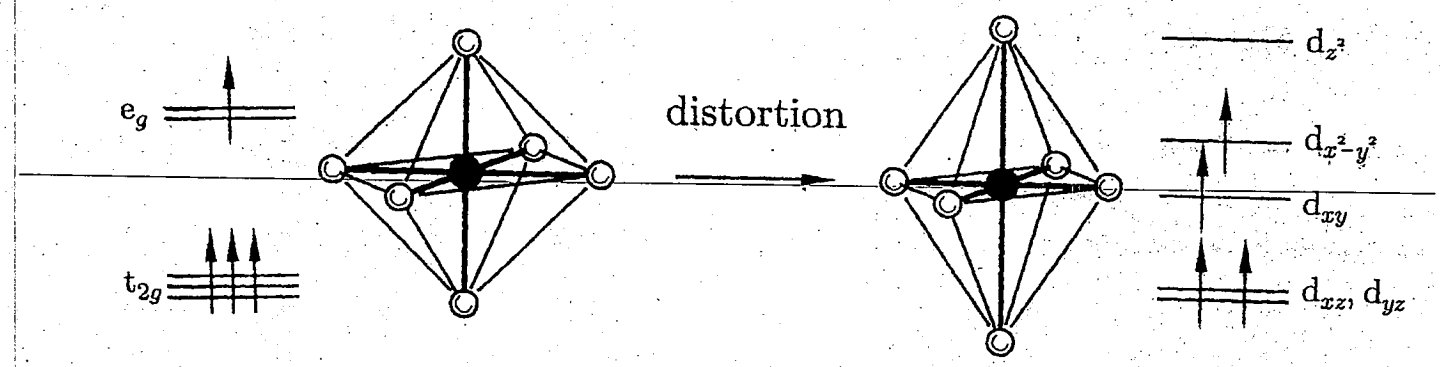
$$\Rightarrow \langle\psi|\vec{L}|\psi\rangle = \int_V \underbrace{\psi^*}_{\text{real}}\vec{L}\psi\, d\tau = \int \psi\vec{L}\psi = \text{imaginary}$$

⇒ real = imaginary ⇒ $\langle\psi|L|\psi\rangle = 0$ ⇒ orbital momentum is quenched!

Coupling to the lattice deformation: Jahn-Teller effect

If orbital ground state is degenerate, then lowering of energy by crystal distortion is possible

Example: e_g electron in a distorded cubic crystal field



Tetragonal distortion removes the degeneracy of e_g electron

Distortion: $\varepsilon \Rightarrow$ elastic energy $E_{el} = A\varepsilon^2$

$\Rightarrow$ variation of crystal field energy $\Delta = B\,\varepsilon$

Minimum of energy: $\varepsilon = -A/2B$

spontaneous distortion; no orbital degeneracy in the ground state